🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.3

Describe Data by Range and Interquartile Range

Lesson 2-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can describe the shape of a data distribution as symmetric, skewed, or having clusters and gaps.

Language Objective: I can explain my description using the words symmetric, skewed, cluster, and gap.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Shape of Data Distributions | Symmetric | Skewed | Cluster | Gap | Data distribution | Variability

👆 *Tap any word to see what it means and a picture.*

1. __________ — The overall pattern of data, such as symmetric, skewed, clustered, or containing gaps.

2. A distribution where both sides look like mirror images is __________.

3. A distribution with a tail stretching out to one side is __________.

4. A group of data values bunched close together is a __________.

5. An interval where there are no data values is a __________.

6 The way data values are spread out across a range is the

____________________ .

7 How much the values in a data set differ from one another is the

____________________ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

DESCRIBE

What is the shape of this distribution? Should the reporter use the mean or median to describe a 'typical' finish time?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Shape of Data Distributions** — The overall pattern of data, such as symmetric, skewed, clustered, or containing gaps.
  Forma de las distribuciones de datos — El patrón general de los datos, como simétrico, sesgado, agrupado o con espacios.
- [ ] **Symmetric** — Data that looks about the same on the left and right.
  Simétrico — Datos que se ven casi iguales a la izquierda y a la derecha.
- [ ] **Skewed** — When most data sits on one side with a tail on the other.
  Sesgado — Cuando la mayoría de los datos está de un lado con una cola del otro.
- [ ] **Cluster** — A group of numbers that are close together.
  Agrupamiento — Un grupo de números que están cerca unos de otros.
- [ ] **Gap** — A big empty space where there is no data.
  Hueco (espacio) — Un espacio grande y vacío donde no hay datos.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The distribution is skewed ____ because the tail stretches toward the ____ finish times. There is a cluster around ____. The reporter should use the ____ because the ____ would be pulled toward the ____ by the extreme values.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Different real-world data sets form different distribution shapes: symmetric, skewed, or clustered.

Evidence: Students explain different data produces different shapes and can describe symmetric (balanced) versus skewed (lopsided) distributions.

Reasoning: This evidence connects the definition of shape of data distributions to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The distribution is skewed ____ because the tail stretches toward the ____ finish times. There is a cluster around ____. The reporter should use the ____ because the ____ would be pulled toward the ____ by the extreme values.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Shape of Data Distributions
2. **Notes 2:** Symmetric
3. **Notes 3:** Skewed
4. **Notes 4:** Cluster
5. **Notes 5:** Gap
6. **Notes 6:** Data distribution
7. **Notes 7:** Variability
8. **Watch for this mistake:** *A common mistake in Shape of Data Distributions is naming the skew by looking at where the TALLEST bar (the peak) sits, instead of looking at which way the TAIL stretches. For example, given bar heights 8, 5, 3, 1 (tall on the left, trailing off to the right), students often say "skewed left" because the peak is on the left — but the correct name is skewed right, because skew is always named for the direction the tail points, not the peak.*
9. **Try It 1:** A. Symmetric with one cluster around the peak — *One peak in the middle with roughly even sides means the data mirrors around the center, so it is symmetric with a single cluster at the peak.*
10. **Try It 2:** A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left — *The data clusters at the high end (70-85%), there is an empty gap (50-65%), and the few low values form a tail on the left, so the display is skewed left.*